Natural Join

$R(X,Y) \quad S(Y,Z)$

insieme di attr. in comune

$R \bowtie S = \{t : t[XY] \in R \text{ AND } t[YZ] \in S\}$

Equi join

R S'

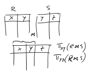
non hanno attrib in comune

θ-Join

$R(X)\ S(Y)$

$X \cap Y = \emptyset$

quando confronto gli attributi non uso =



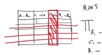
$\Pi_{XY}(R \bowtie S) \subseteq R$

$\Pi_{YZ}(R \bowtie S) \subseteq S$

$R(A_1, A_2 \ldots A_n, C_1, C_2 \ldots C_k)$ $n+k$

$S(C_1, C_2 \ldots C_k, B_1, B_2 \ldots B_m)$ $m+k$

$R \bowtie S \quad (A_1 \ldots A_n, C_1 \ldots C_k, B_1 \ldots B_m)$



$\Pi_{A_1 - A_n, C_1 - C_k, B_1 - B_m}\left(\sigma_{\substack{C_1 = C'_1 \text{ AND } C_2 = C'_2 \ldots \text{ AND } C_k = C'_k}}\left[R \times S_{\substack{C_1 \to C'_1 \\ C_2 \to C'_2 \\ \vdots \\ C_k \to C'_k}}(S)\right]\right)$

$= R \bowtie S$

Ridenominazione

Dipendente(matricola, nome, stipendio)

matr	nome	stip
123	Pippo	100
234	Pluto	200
456	Paperino	300
:	:	:

$\text{Dipendenti} - \Pi_{\text{nome, stip, matr}}\left[\sigma_{\text{stip} < s}\left[\text{Dipendenti} \times \underbrace{\delta_{\substack{\text{matr} \to m \\ \text{nome} \to n \\ \text{stip} \to s}}(\text{Dipendenti})}\right]\right]$

	matr	nome	stip	m	n	s
	~~123~~	~~Pippo~~	~~100~~	~~123~~	~~Pippo~~	~~100~~
✓	123	Pippo	100	234	Pluto	200
✓	123	Pippo	100	456	Pap.	300
	~~234~~	~~Pluto~~	~~200~~	~~123~~	~~Pippo~~	~~100~~
	~~234~~	~~Pluto~~	~~200~~	~~234~~	~~Pluto~~	~~200~~
✓	234	Pluto	200	456	Pap	300
	~~456~~	~~Pap~~	~~300~~	~~123~~	~~Pippo~~	~~100~~
	~~456~~	~~PAP~~	~~300~~	~~234~~	~~Pluto~~	~~200~~
	~~456~~	~~PAP~~	~~300~~	~~456~~	~~Pap~~	~~300~~

⟹

matr	nome	stip
123	Pippo	100
234	Pluto	200

$\text{Dipendenti} -$

$\Pi_{\substack{\text{matr} \\ \text{nom} \\ \text{stip}}}\left(\text{Dipend} \underset{\text{stip} < s}{\bowtie} \delta_{\substack{\text{matr} \to m \\ \text{nome} \to n \\ \text{stip} \to s}}(\text{Dipend})\right)$

Output

nom	matr	stip
Pap	456	300

Dipendenti(matr, nome, stip, reparto)

Per ogni reparto trovare e visualizzare lo stip più alto

$R \bowtie S \equiv$

$X \cap Y = \emptyset$

$R \times S$

Studenti(matr, nome)

Esame(matr, materia, voto)

$\text{Studenti} \;\boxed{\bowtie}\; \text{Esame} =$

$(\text{Studenti} \bowtie \text{Esame}) \cup$

$\underbrace{\left(\text{Studenti} - \Pi_{\substack{\text{matricola,} \\ \text{nome}}}(\text{Studenti} \bowtie \text{Esame})\right)} \times \{\text{materia} = \text{NULL}, \text{voto} = \text{NULL}\}$

$R := \sigma_{\text{eta} > 40}(\text{Persone})$

$t \in R$

S $\quad S \times \{t\}$

Esami(matricola, corso)

Corsi(corso)

$R(X,Y)$

$S(Y)$

$R \div S = \{w \mid \{w\} \times S \subseteq R\}$

$\{123\} \times \text{Corsi}$

m	c
123	DB
123	Prog

?

$\{234\} \times \text{Corsi}$

m	c
234	DB
234	Prog

$\{456\}$

Esami

matr	corso
123	DB
123	Prog
234	DB
456	Prog

Corsi

corso
DB
Prog

$\text{Esami} \div \text{Corsi}$

matr
123

Prodotto(codice, nome, descr, tipo)

cliente(CF, nome)

Acquisto(CF, codice, data)

Q: trova i clienti che hanno acquistato tutte le tipologie di prodotto !!

$R := \Pi_{\substack{CF, \\ \text{tipo}}}(\text{Acquisto} \bowtie \text{Prodotto})$

$S := \Pi_{\text{tipo}}(\text{Prodotto})$

$R(CF, \text{tipo}) \qquad S(\text{tipo})$

$R \div S$